TUNDUMA TOWN COUNCIL

SUMAGHA SECONDARY SCHOOL

FORM TWO ASSESSMENT SERIES I

037 COMPUTER SCIENCE -- MARKING SCHEME

Year: 2026

SECTION A (15 Marks)

1. Multiple Choice (10 Marks, 1 mark each)

Q	Answer
(i)	D - UPS
(ii)	B - To restrict changes of files done by the user
(iii)	C - Microprocessor
(iv)	A - 100 (10>=20 is False, so first*10 = 100)
(v)	B - Spreadsheets
(vi)	A - Ctrl + S
(vii)	D - Google mail
(viii)	C - int number = 5;
(ix)	B - Netiquette
(x)	D - To make typing easier

2. Matching (05 Marks)

Item	Answer
(i)	B - Compiler
(ii)	D - Interpreter
(iii)	A - Assembler
(iv)	F - Translator
(v)	E - Source code

SECTION B (70 Marks)

3(a) Three institutions (03 Marks)

- Hospital
- Bank
- School (accept any reasonable institution, e.g. police station, factory, government office)

3(b) One function each (06 Marks)

- Hospital - used to store and retrieve patients' medical records and for diagnosis
- Bank - used to process transactions and keep customers' account records
- School - used to keep students' records and to support teaching and learning

3(c) Component expected (01 Mark)

Marking guide: Any hardware component e.g. Monitor / System unit / Keyboard - accept one reasonable answer.

4(a) Concept of internet (02 Marks)

Marking guide: The internet is a global network of interconnected computer networks that communicate using standard protocols (TCP/IP) to share information and resources worldwide.

4(b) Three uses of internet at school (06 Marks)

- Research - students search for information for assignments and projects
- Communication - emailing teachers/students, video conferencing
- E-learning - accessing online lessons, notes and learning materials

4(c) Two email software (02 Marks)

- Gmail
- Microsoft Outlook (accept Yahoo Mail etc.)

5(a) Types of software (03 Marks)

- Windows 10 - System software (Operating System)
- Office 2016 - Application software
- Avast antivirus - Utility software

5(b) Function of each type (06 Marks)

- Operating System - manages hardware and software resources and provides the interface between the user and the computer
- Application software - helps the user perform specific tasks, e.g. word processing, spreadsheets
- Utility software (antivirus) - protects the computer from viruses and malware

5(c) Why installed on hard disk (01 Mark)

Marking guide: Because the hard disk is non-volatile storage, so software remains saved permanently even when the computer is switched off.

6(a) Five factors for a strong password (05 Marks)

- Use a mix of uppercase and lowercase letters
- Include numbers
- Include special characters/symbols
- Make it at least 8 characters long
- Avoid using personal information such as name or birth date

6(b) Five components of cybersecurity (05 Marks)

- Network security - protects computer networks from intruders
- Application security - keeps software and devices free of threats
- Information security - protects the integrity and privacy of data
- Operational security - covers processes and decisions for handling data assets
- Disaster recovery / business continuity - plans how an organisation responds to a cyber incident

7(a) Five reasons to keep software updated (05 Marks)

- Fixes security vulnerabilities

- Improves performance
- Adds new features
- Fixes bugs/errors
- Ensures compatibility with new hardware/software

7(b) Three steps to remove malware (03 Marks)

- Run an antivirus/anti-malware scan
- Delete or quarantine the infected files
- Restart the computer and update the antivirus software

7(c) Two preventive maintenance conditions (02 Marks)

- The computer must be switched off and disconnected from power
- Proper tools and antistatic precautions must be used

8(a) Variable vs Array (02 Marks)

Marking guide: A variable stores a single value at a time, while an array stores multiple values of the same data type under one name, accessed by index.

8(b) C program - sum, difference, quotient, remainder (04 Marks)

```
#include <stdio.h>
int main() {
int a, b;
printf("Enter two numbers: ");
scanf("%d %d", &a, &b);
printf("Sum = %d\n", a + b);
printf("Difference = %d\n", a - b);
printf("Quotient = %d\n", a / b);
printf("Remainder = %d\n", a % b);
return 0;
}
```

8(c) C program - array of 5 numbers (04 Marks)

```
#include <stdio.h>
int main() {
int nums[5] = {1, 2, 3, 4, 5};
int i;
for (i = 0; i < 5; i++) {
printf("%d\n", nums[i]);
}
return 0;
}
```

9(a) Running in Thonny (02 Marks)

Marking guide: Press F5 (or click the Run button) to run the program. Expected output: Jambo, Tanzania!

9(b) Voting eligibility program (04 Marks total)

```
age = int(input("Enter your age: "))
citizen = input("Are you a citizen? (Yes/No): ")
if age < 0:
print("Invalid age")
elif age >= 18 and citizen == "Yes":
print("Eligible to vote")
else:
print("Not eligible to vote")
```

9(c) Print numbers 1-10 (02 Marks)

```
for i in range(1, 11):
print(i)
```

9(d) welcome_message function (02 Marks)

```
def welcome_message(name):
print("Welcome, " + name + "!")

welcome_message("John")
```

SECTION C (15 Marks)

10. Six main aspects of troubleshooting (15 Marks - 2.5 marks each)

- Identify the problem - gather information about the fault reported
- Establish a theory of probable cause - form a hypothesis of what may be wrong
- Test the theory to determine the actual cause
- Establish a plan of action to resolve the problem and implement the solution
- Verify full system functionality after the fix
- Document findings, actions and outcomes for future reference